

Design and Innovation Processes: Scripts and Configuring Users

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About Me

- Doctoral Candidate in Technology & Health
- Fields: Science & Technology Studies (STS); AI Ethics; Human-Computer Interaction (HCI); Ageing; Sociology of Health
- Methods: Qualitative, Thematic Analysis, Participatory Design (HCI), User-centered approaches (STS), AI development, Policy Analysis
- Education: BSc. Mathematics (Dublin City University), MSc. Machine Learning (KTH), PhD. Technology & Health



Agenda for Today

13:15-14:00

- Introduction to STS, socio-technical imaginaries (Jasanoff) and semiotic approaches to users (15 mins)
- Discuss in groups: Semiotic approaches to users (10 mins)
- Introduction to scripts and configuring users (5 mins)
- Group work: Apply concepts to technologies (15)

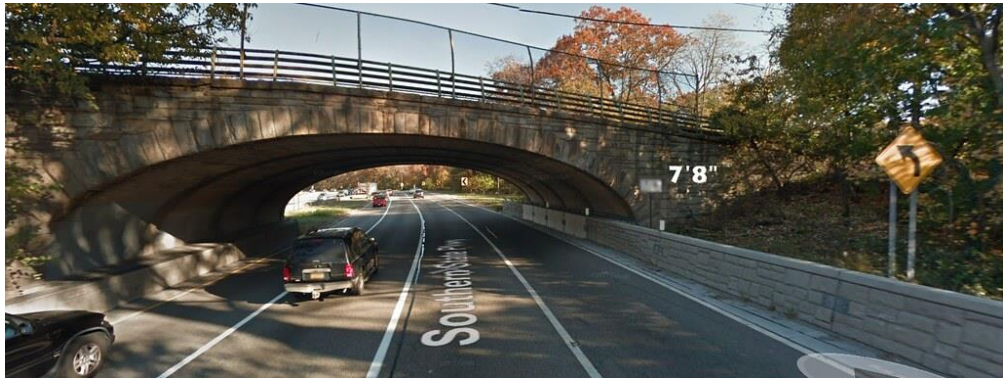
14:15-15:00

- Present slides from group work (10 mins)
- Introduction to Social construction of technology (10 mins)
- Group work: Discuss comparison of theories (15 mins)
- Conclusions (10 mins)

Low Overpass Bridges

- Low overpass bridges (e.g. on Long Island Parkways in New York) were deliberately built too low for buses to pass.
- Prevented buses (and therefore poorer, often Black communities) from accessing certain areas like beaches
- Favoured car owners (typically wealthier groups)
- Even something as “technical” as bridge height can encode **social exclusion**.
- Technologies and infrastructures can **embed political decisions**





Source: Langdon Winner – Do Artifacts Have Politics?

Nuclear Reactor Design Choices

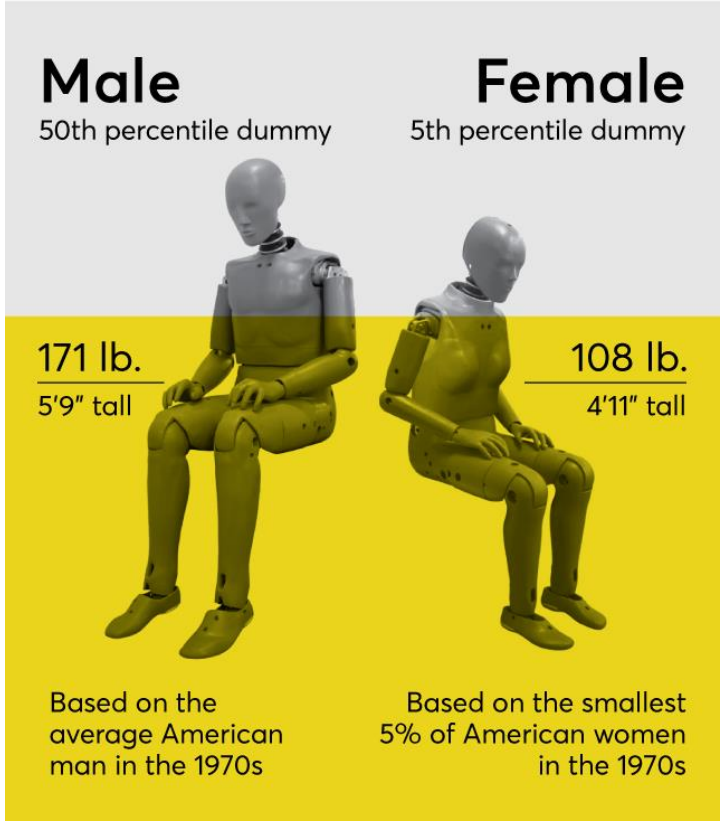
- Langdon Winner: “The increased deployment of nuclear power facilities must lead society toward authoritarianism”
- Some technologies (like nuclear power) require centralieed control structures. Others (like solar) can be decentralized
- Nuclear is not easily compatible with decentralised or fully democratic control → concentrates power in governments and technical elites
- Technologies can favour certain political systems



Source: Langdon Winner – The Whale and The Reactor

Seatbelt Design

- Safety testing and evaluation historically used male crash-test dummies
- Women are more likely to be killed (17%) and injured (73%) in crashes
- Seatbelts do not fit different body types equally
- Design choices reflect social assumptions (who is the “default” user)



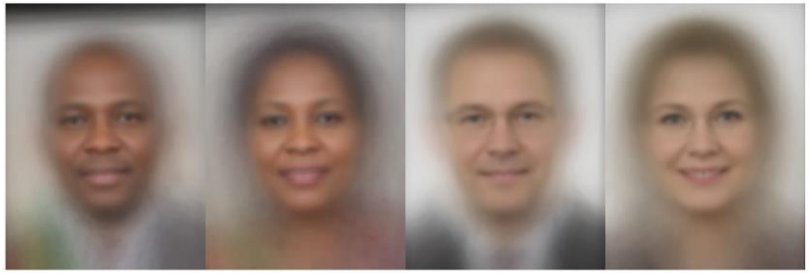
Source: Consumer Reports – The Crash Test Bias: How Male-Focused Testing Puts Female Drivers at Risk (2019)

Facial Recognition Technology

- Enables: security & policing, surveillance, social control, reduced privacy
- Algorithms reflect training data and developer choices
- Facial recognition technology reflects socia values and reinforces power structures
- It is socially constructed (SCOT) and politically impactful (Winner)



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



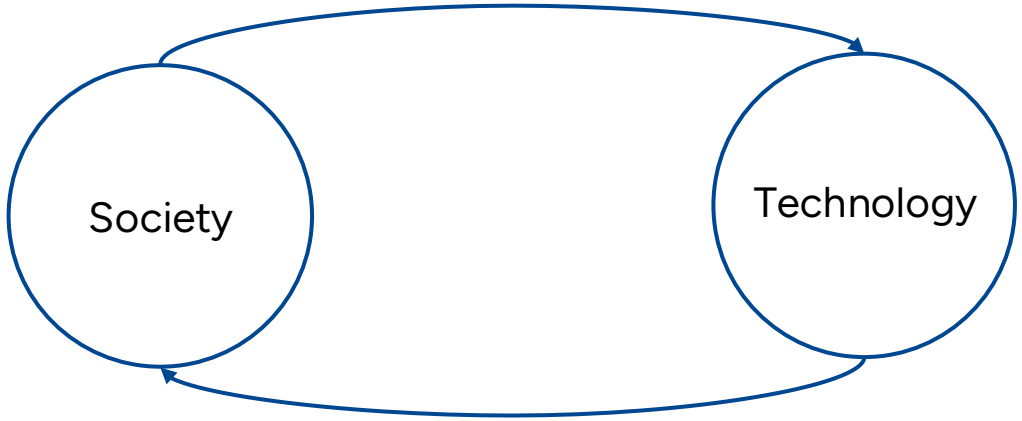
Source: Joy Buolamwini, Gender Shades project

TECHNOLOGY IS NOT VALUE-NEUTRAL!



Science & Technology Studies (STS)

- STS examines the complex interrelation between science, technology, and society
- It **challenges the view of technology as an autonomous force** that develops according to its own internal logic and subsequently impacts society in predetermined ways



- STS emphasizes that technologies are fundamentally social products, shaped by the contexts in which they are designed, developed, deployed, and used (mutual shaping).
- Today, we ask important questions about the role of the designer!

Socio-technical Imaginaries

How do shared visions of science & technology shape societal expectations?

Actively shape research agendas, guide investment and governance

"Collectively held, institutionally stabilised, and publicly performed **visions of desirable futures**, animated by shared understandings of forms of social life and social order attainable through, and supportive of, advances in science and technology"
-Jasanoff & Kim

Studying **emerging technologies** at moments of openness, before their meanings and uses become fully fixed (Borup 2006)

Influence how technologies are designed, implemented, and interpreted in practice

Source: Jasanoff, Sheila, and Sang-Hyun Kim, eds. Dreamscapes of modernity: Sociotechnical imaginaries and the fabrication of power. University of Chicago Press, 2019.
Borup, Mads, et al. "The sociology of expectations in science and technology." Technology analysis & strategic management 18.3-4 (2006): 285-298.

Recap

- Sociotechnical imaginaries show **how society collectively envisions desirable technological futures**.

- These visions influence **who is imagined as a user** and **how technologies are designed to shape user behavior**.

- Semiotic approaches zoom in on the **micro-level of technology design**, focusing on the **scripts, inscriptions, and user representations embedded in artifacts**.

- From collective visions → concrete user configurations in devices, software, and systems.

Semiotic Approaches to Users

- "Semiotics" → the study of signs and symbols, what they mean, and how they are used
- In STS, semiotics is about how artifacts "communicate" instructions, norms and assumptions to users
- Technology "talks" to users via scripts and inscriptions.
- Different designs imagine different types of users (expert, novice, passive, active).

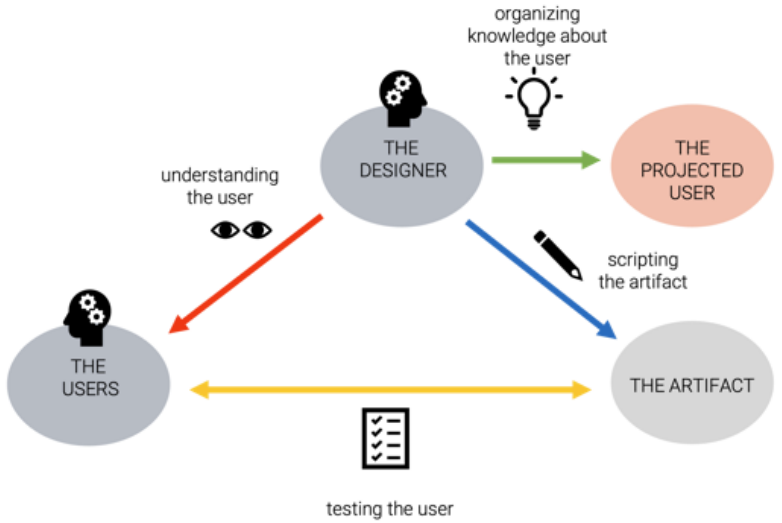


Figure 8: The user-artifact-designer relationship centered around the designer.

Source: Araujo de Aguiar, Carlos Henrique, Trevor Pinch, and Keith Green. "De-scription at early phases of artifact design." Proceedings of the 25th International Academic Mindtrek Conference. 2022.

Discuss in Groups (10 mins)

- What are the main concepts of semiotic approaches to users?
- How are they related to each other?
- Collect your ideas on Google doc: link
- You can find resources in the preparatory readings: (Oudshoorn and Pinch 2003; Latour 1991; Woolgar 1991; Akrich 1992; Akrich 1995)



User Representation & Representation Techniques

- **User representations:** the process by which innovators and designers construct visions of their future users and integrate these representations into technical choices. This "inscribed" vision of the world within a technical object is referred to as a "script" or a "scenario"

- **Representation techniques:** refer to different ways engineers and designers may seek to develop user representations
  - Include implicit and explicit representation techniques
  - Explicit: customer testing, feedback on experience, market surveys
  - Implicit: existing products, consulting experts, reliance on personal experience

Source: Akrich, M (1995) User representations: Practices, methods and sociology. In: Rip, A, Misa, TJ, Schot, JW (eds) Managing Technology in Society. The Approach of Constructive Technology Assessment. London: Pinter, 167–184.

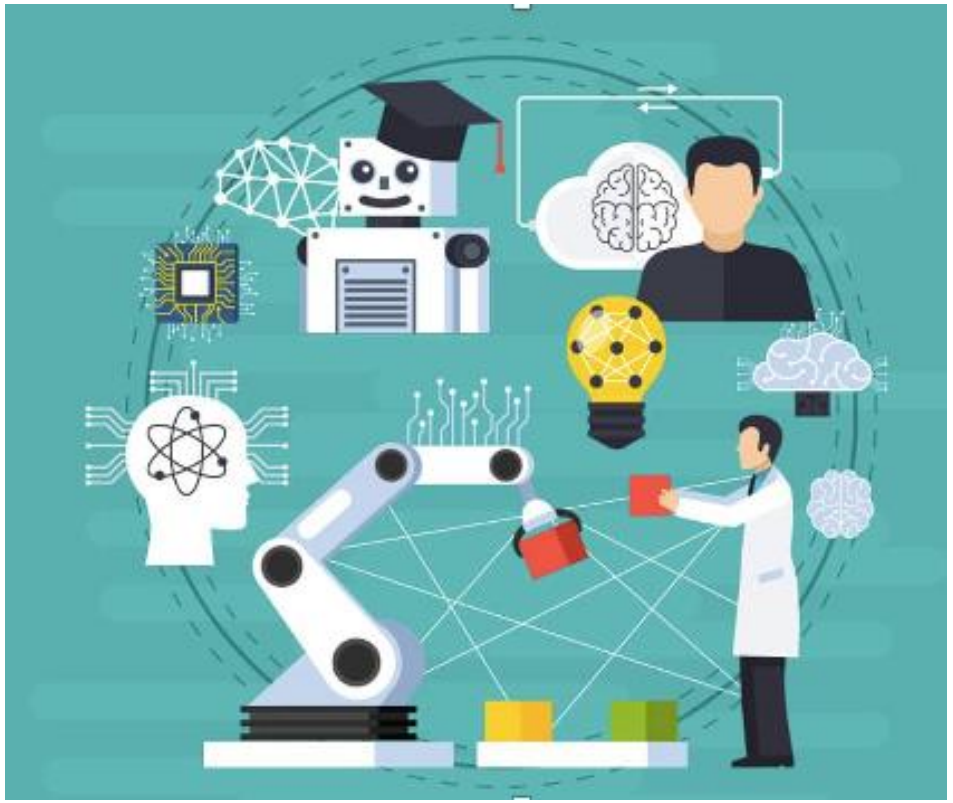
Inscriptions, Scripts, Configuring the User

- **Inscription:** Humans and nonhumans may implicitly encode particular scripts into things
- **Script:** (or scenario) is the end product of the inscription process. Much like a film script, technical objects define a framework of action for the actors and the space in which they are supposed to act
- **Configuring:** By including such particular scripts, machines may directly or indirectly configure user's behaviour
- **De-inscription (De-scription):** Akrich uses the term "de-scription" to refer to the analytical inventory and analysis of the mechanisms that allow the relationship between a technical object's form and its meaning to come into being. It is the process of "reading" the script back out of the object

Sources: Latour and Woolgar 1979; Latour 1991; Akrich 1991; Woolgar 1991

Discuss in Groups (10 mins)

- Select a particular technology you have been interested in.
- Discuss how the concepts discussed so far in class would apply to this technology
- Collect your ideas on Google doc: link
  - What collective visions are embedded?
  - What kind of scripts and user representations are encoded in this technology?
  - How do these influence the user's behaviour?
  - Be prepared to select one person to present your findings to the other groups at the end of the discussion

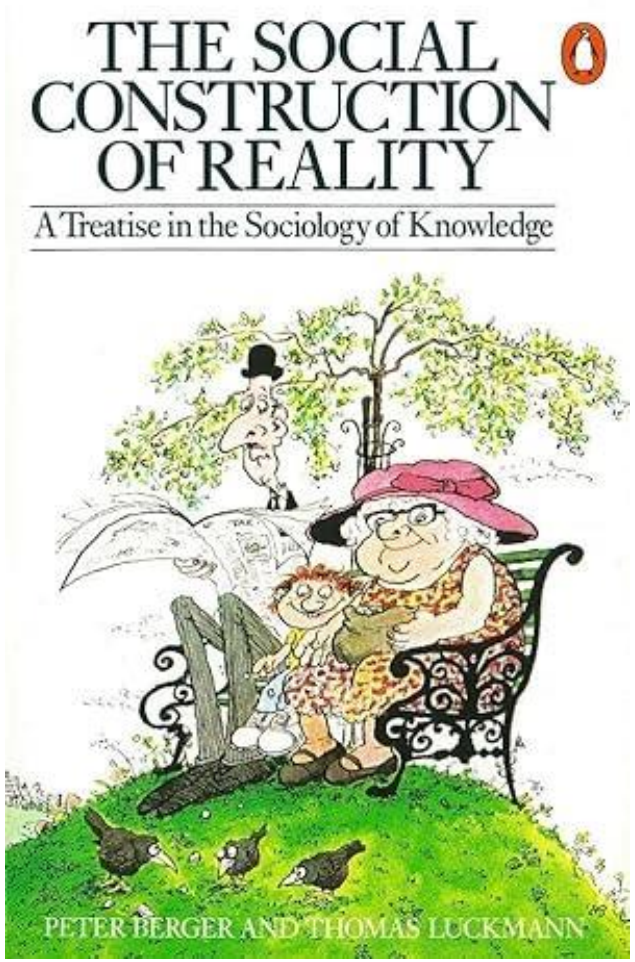


BREAK!



Social Constructivism

- Sociological theory of knowledge according to which human development is socially situated
- Knowledge is constructed through interaction with others
- Facts depend on the shared ways of thinking about and representing the world that groups of people develop collaboratively
- Institutionalization: When these roles are made available to other members of society, the reciprocal interactions are said to be institutionalized.
- Knowledge and people's conceptions (and beliefs) of what reality is become embedded in the institutional fabric of society



Sources: Berger & Luckmann 2002

Social Constructivism

- Facts are constructed within language
- Each person and group creates their own meaning of the reality
- Facts are representations, rather than physical entities
- Facts are institutionalised: embedded in social systems, social networks, organisations and society

→ STS approach: Social Construction of Technology (Pinch & Bijker 1984)

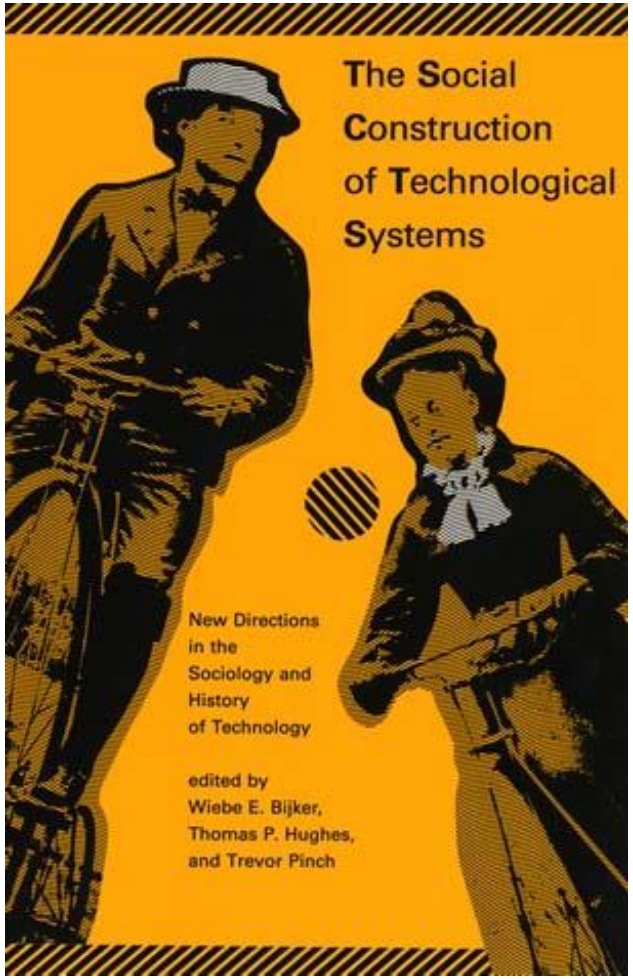


Social Construction of Technology

- **Relevant Social Groups:** Groups of individuals that share a particular perspective on the content of an artefact

- **Interpretive Flexibility:** Not only flexibility in how people interpret technologies, but also flexibility in how technologies are designed. Technologies do not have fixed meanings but are understood differently by different actors

- **Closure & Stabilization:** Over time, the different relevant social groups are negotiating their various interpretations and designs until they achieve a consensus; and one particular technology is stabilized



Discuss in groups (10 min)

- How could you apply SCOT in your project? What are potential strengths and weaknesses of the theory?
- How could you apply semiotic approaches to users in your project? What are potential strengths and weaknesses to the theory?
- Make notes, and be prepared to select one person to outline the main findings of your discussion.



Conclusion

Socio-technical Imaginaries

- Shared visions of desirable futures shaped by society and technology
- Guide innovation, research, and policy priorities
- Reflect cultural values, norms, and expectations
- Mediate between technological possibilities and social goals
- Highlight collective imaginaries rather than individual choices
- Limitation: May overlook individual or cross-community differences, especially at larger scales

Semiotic Approaches to Users

- Focus on networks linking humans and technologies
- Reveal power differences between designers and users
- Technology partially shapes society (weak technological determinism)
- Emphasises how technology enables or constrains user action
- Shows how meanings and interpretations are embedded in design
- Limitation: Struggles to fully capture user agency and long-term innovation
- How users and designers interact? → Lecture 4

Social Construction of Technology

- Technology development shaped by socio-historical processes
- Emphasises user and engineer interpretations of technology
- Society largely informs technological outcomes (social determinism)
- Highlights negotiation and contestation over meanings
- Accounts for flexibility and multiple potential designs
- Limitation: Less attentive to power structures and broader societal forces